

2016.0 RANGE ROVER (LG), 501-29

SIDE PANEL SHEET METAL REPAIRS

ROCKER PANEL (G1454630)

REMOVAL AND INSTALLATION

REMOVAL

NOTES:

- Where Self Piercing Rivets (SPR's) removal is instructed use either the approved SPR tool or 5.3mm drill bit
- Where Breakstem Fasteners (BSF) removal is instructed remove fastener shaft with a 4mm punch and remove the head with a 6.5mm drill bit

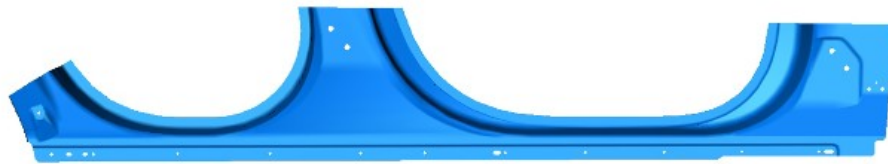
1.

NOTE:

The removal and installation steps shown within this procedure may differ depending on whether the panel in question is being replaced in isolation, or, is being replaced in combination with panels other than those listed.

The rocker panel is a category A repair.

2.



E143359

The rocker panel is serviced as a separate panel.

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3. For additional information, refer to:
[Health and Safety Precautions](#) (100-00 General Information, Description and Operation),
Corrosion Prevention (501-25 Body Repairs - General Information, Description and Operation),
[Body and Frame](#) (501-26 Body Repairs - Vehicle Specific Information and Tolerance Checks, Description and Operation).

 4. Disconnect the battery ground cable,
For additional information, refer to: [Battery](#) (414-01 Battery, Mounting and Cables, Removal and Installation).

 5. Remove the front seat,
For additional information, refer to: [Front Row Seat](#) (501-10B Seating - Long Wheel Base, Removal and Installation).

 6. Remove the rear seat cushion,
For additional information, refer to: [Rear Seat Cushion](#) (501-10B Seating - Long Wheel Base, Removal and Installation).
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7. Remove the rocker panel moulding,
For additional information, refer to: [Rear Rocker Panel Moulding - Long Wheel Base](#) (501-08 Exterior Trim and Ornamentation, Removal and Installation).

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8. Remove the front and rear wheels from the damaged side of the vehicle,
For additional information, refer to: [Wheel and Tire - GTDi 2.0L Petrol/TDV6 3.0L Diesel /TDV8 4.4L Diesel/V6 S/C 3.0L Petrol /V8 N/A 5.0L Petrol/V8 S/C 5.0L Petrol](#) (204-04 Wheels and Tires, Removal and Installation).

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9. Remove the front and rear fender splash shield,
For additional information, refer to: [Rear Wheel Arch Liner](#) (501-08 Exterior Trim and Ornamentation, Removal and Installation).

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10. Remove the scuff plate trim panel,
For additional information, refer to: [Rear Scuff Plate Trim Panel - Long Wheel Base](#) (501-05 Interior Trim and Ornamentation, Removal and Installation).

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11. Release the floor carpet and position it to one side.

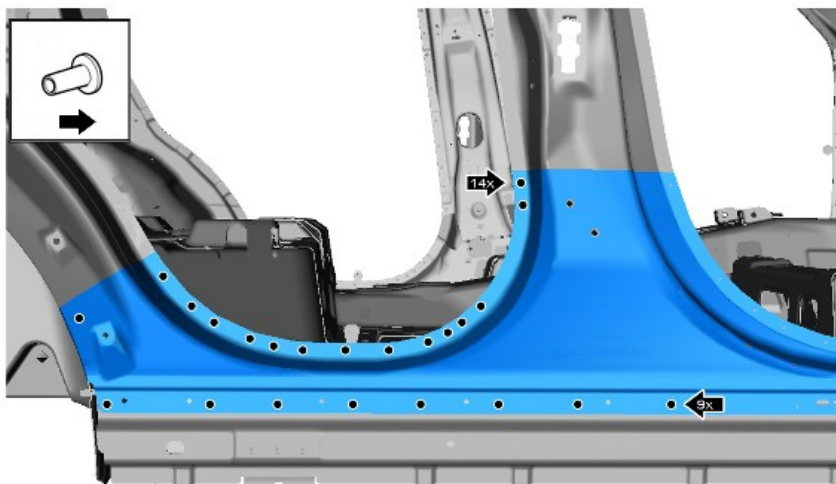
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12. **NOTE:**

If the front fender lower mounting bracket is undamaged, retain for re-use on installation.

Remove the front fender lower mounting bracket,
For additional information, refer to: [Front Fender Lower Mounting Bracket](#) (501-27 Front End Sheet Metal Repairs, Removal and Installation).

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13. Release the rocker panel wiring harness and position it to one side.

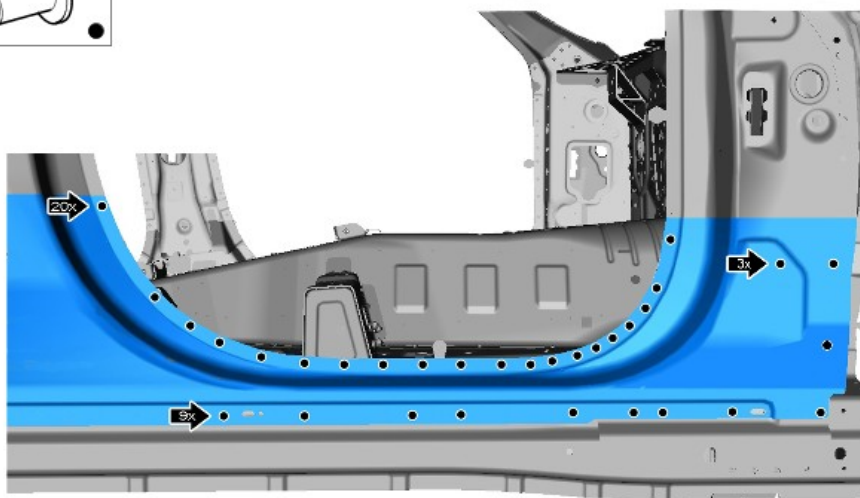
14.



E143360

Remove the SPR's.

15.



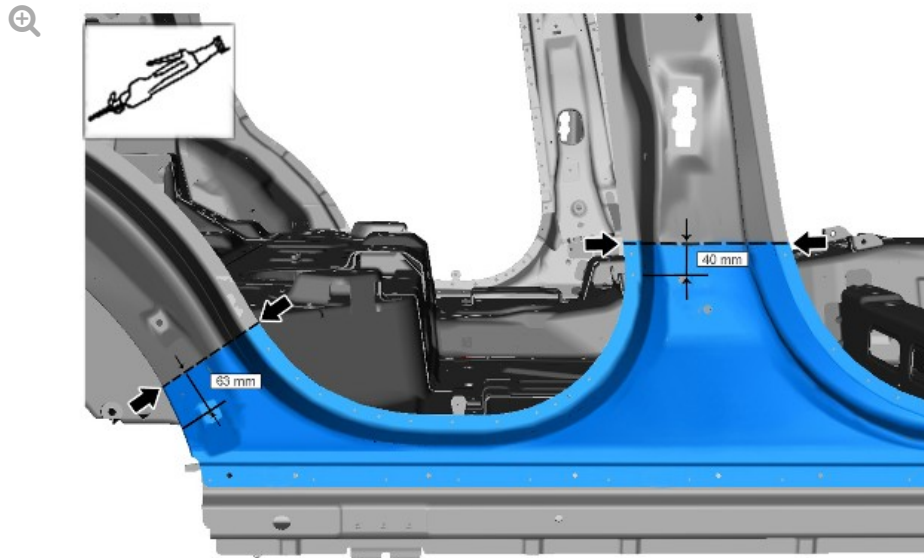
E143361

Remove the SPR's.

16.

CAUTION:

Care should be taken not to cut through into inner panels or reinforcements.



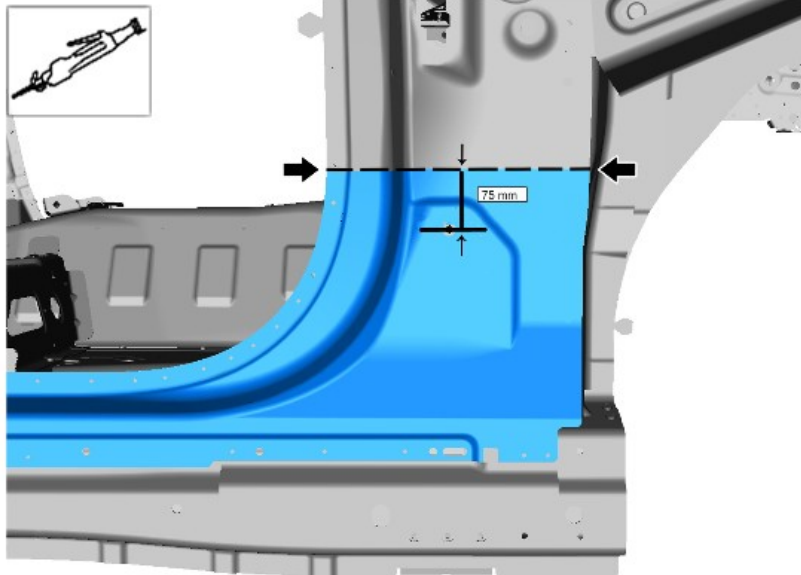
E143362

Mark out the positions where the rocker panel MIG (metal inert gas) butt joint is to be made. Cut through the rocker panel at this point, as indicated.

17.

NOTE:

Care should be taken not to cut through into inner panels or reinforcements.

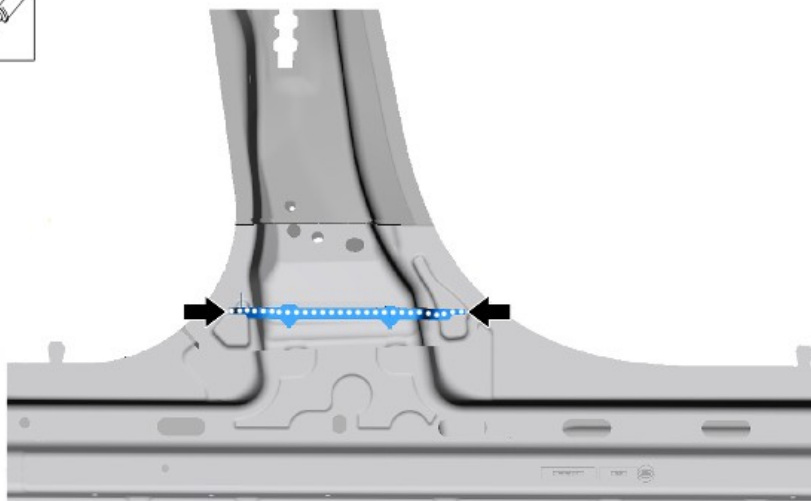


E143363

Mark out the position where the rocker panel MIG butt joint is to be made. Cut through the rocker panel at this point, as indicated.

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18. Separate the joints and remove the rocker panel
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19.



E143377

Release the NVH (noise, vibration and harshness), components.

INSTALLATION

CAUTION:

Make sure any panel joints containing adhesive are fully clamped to avoid any separation prior to the installation of external panels. These joints must be left clamped until the adhesive has cured

NOTES:

- Breakstem Fastener (BSF) installation involves using the Land Rover approved BSF installation tool and requires a hole diameter of 6.5mm
- Self Piercing Rivet (SPR) installation requires using the Land Rover approved SPR tool

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1. Remove rivet remnants.
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2. Dress flanges where necessary.
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3. Using a fine bristle disc, clean and prepare the panel surfaces, removing any sealer and adhesive residue.
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4. Offer up the new panel and clamp into position. Check alignment, if correct proceed to next step, if not, rectify and recheck before proceeding.
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5.

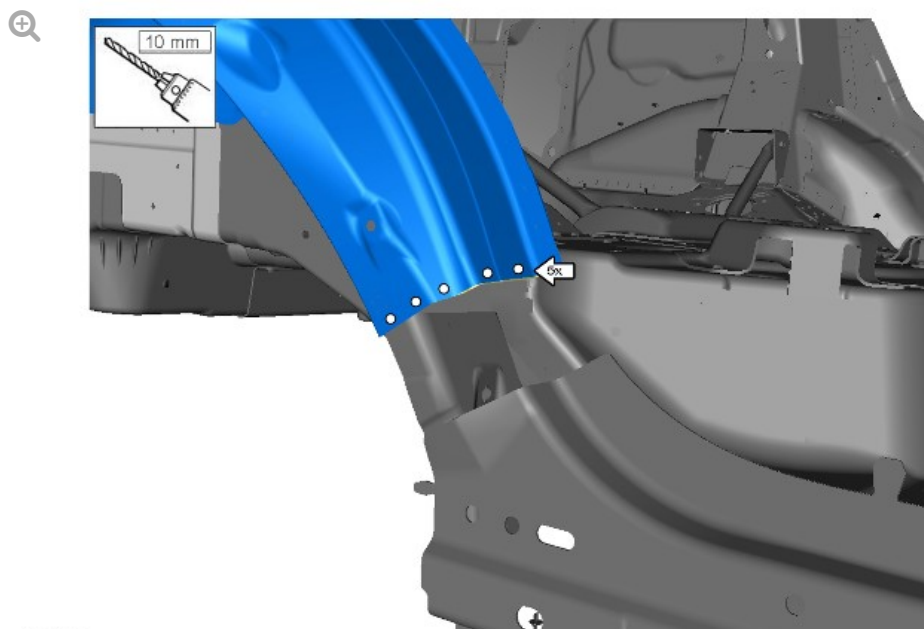


E143364

Drill holes through the old panels into the new panel, at the points where BSF's are to be installed.

6. Remove the new panel.

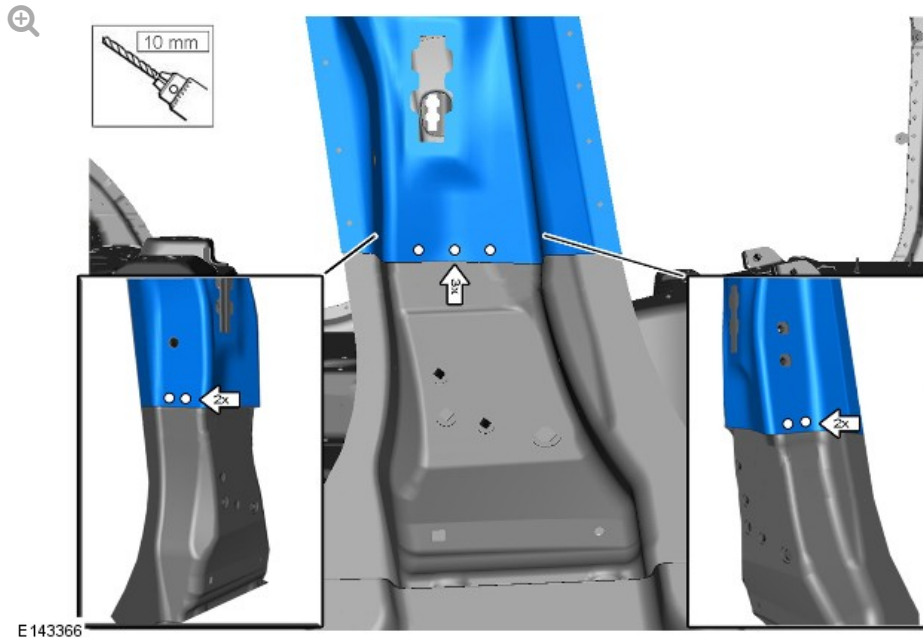
7.



E143365

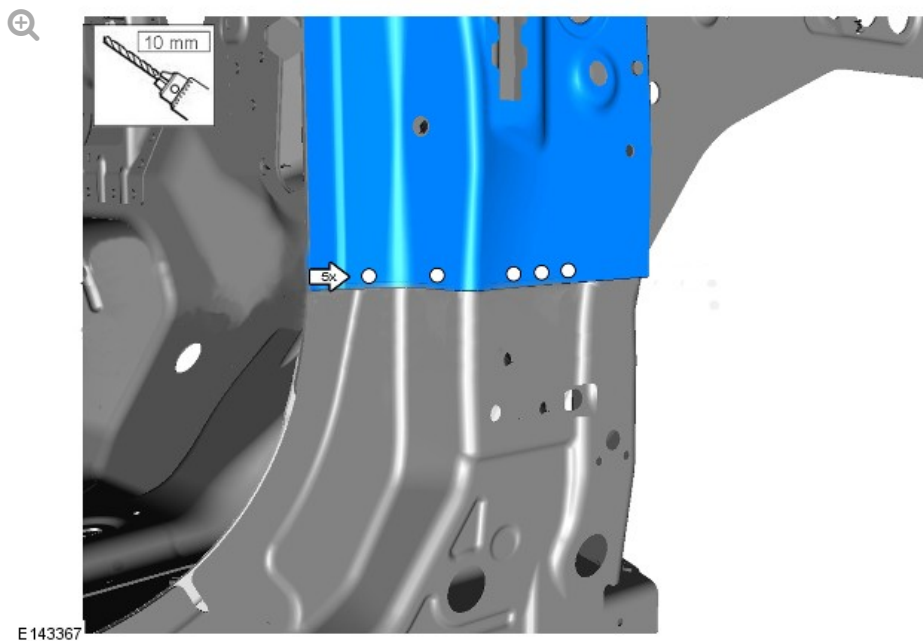
Using a 10mm drill bit, drill holes for MIG plug welds for the installation of the backing plate as indicated.

8.



Using a 10mm drill bit, drill holes for MIG plug welds for the installation of the backing plate as indicated.

9.



Using a 10mm drill bit, drill holes for MIG plug welds for the installation of the backing plate as indicated.

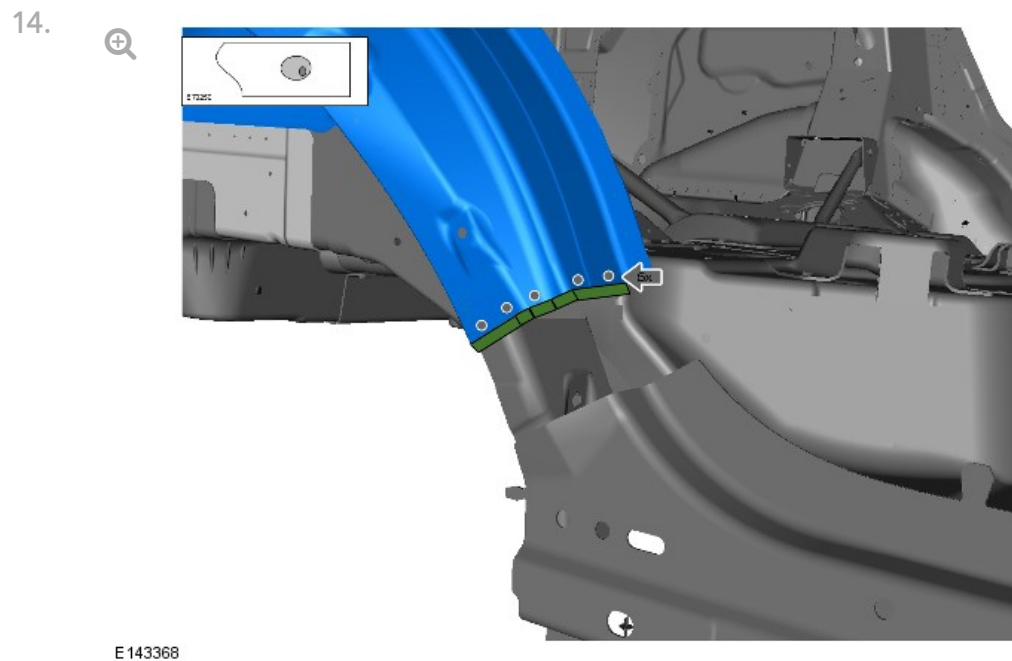
10. Deburr the drilled holes.

11. Measure and cut three 50mm wide backing plates, 25mm each side

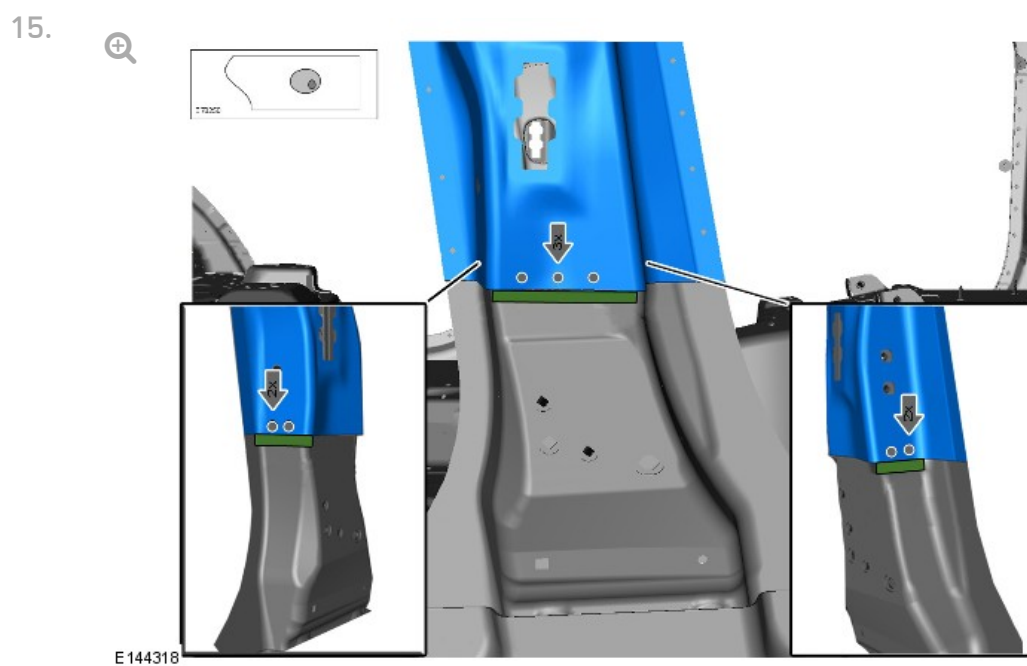
of the MIG butt joint, from the discarded part of the rocker panel.

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12. Cut six run on/run off tabs from the discarded part of the rocker panel.
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13. Using a fine bristle disc, clean and prepare the panel surfaces.
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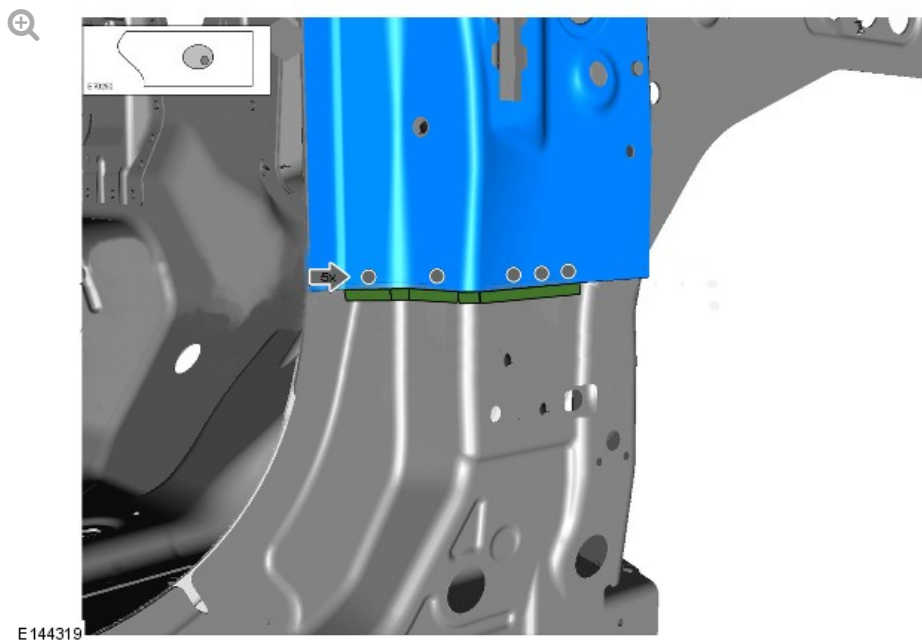


MIG plug weld the backing plate as indicated.



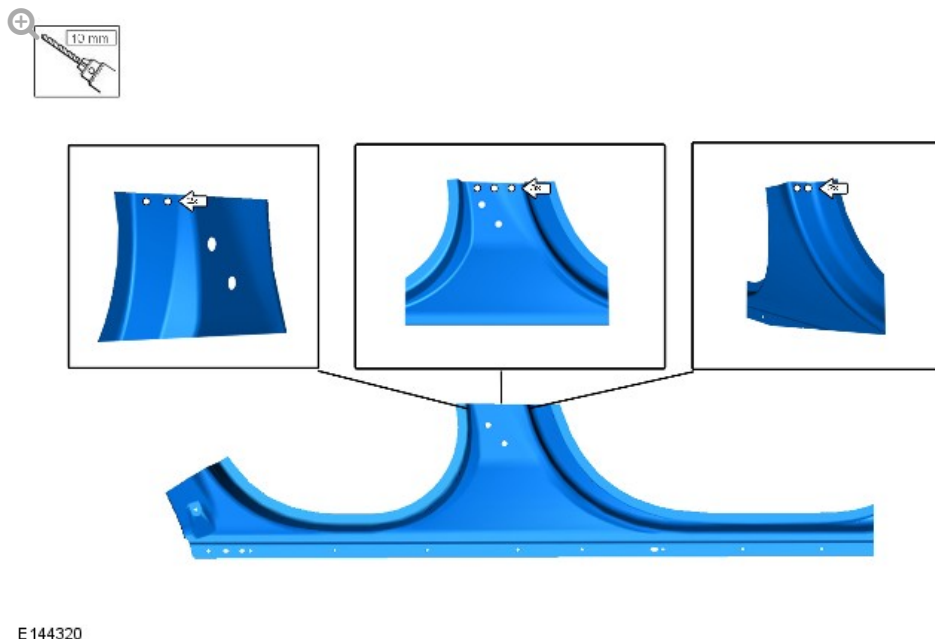
MIG plug weld the backing plate as indicated.

16.



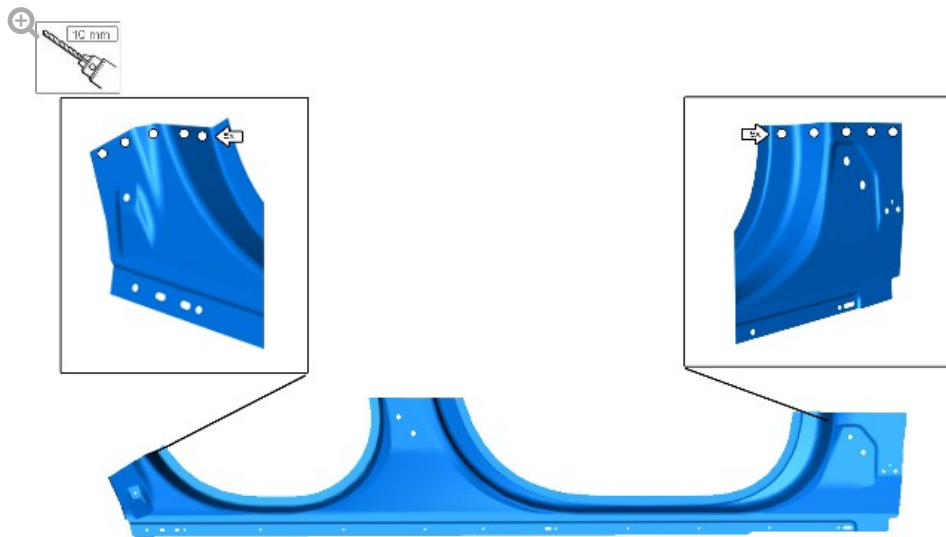
MIG plug weld the backing plate as indicated.

17.



Using a 10mm drill bit, drill holes for MIG plug welds for the installation of the backing plate as indicated.

18.

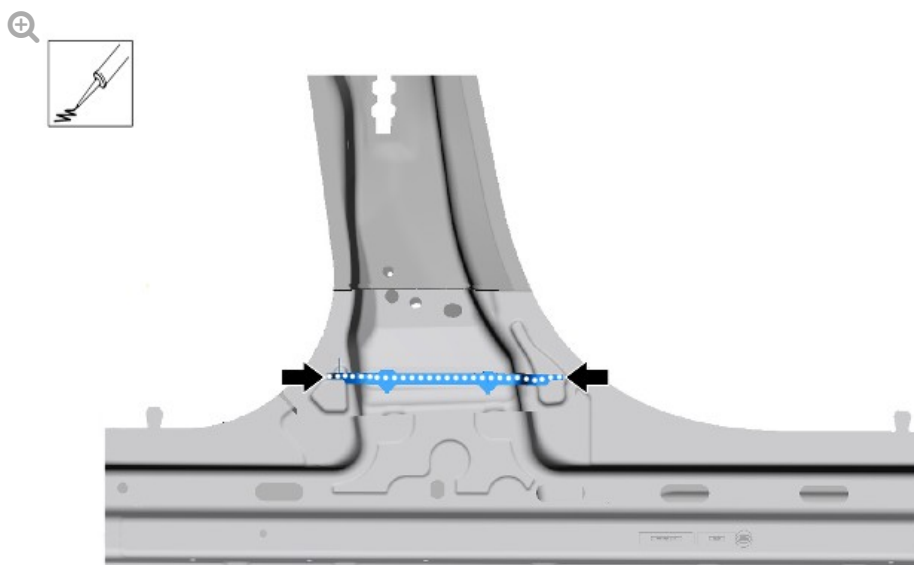


E144321

Using a 10mm drill bit, drill holes for MIG plug welds for the installation of the backing plate as indicated.

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19. Apply the coupling agent where the Land Rover recommended bonding material is to be applied and allow to dry.
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20.



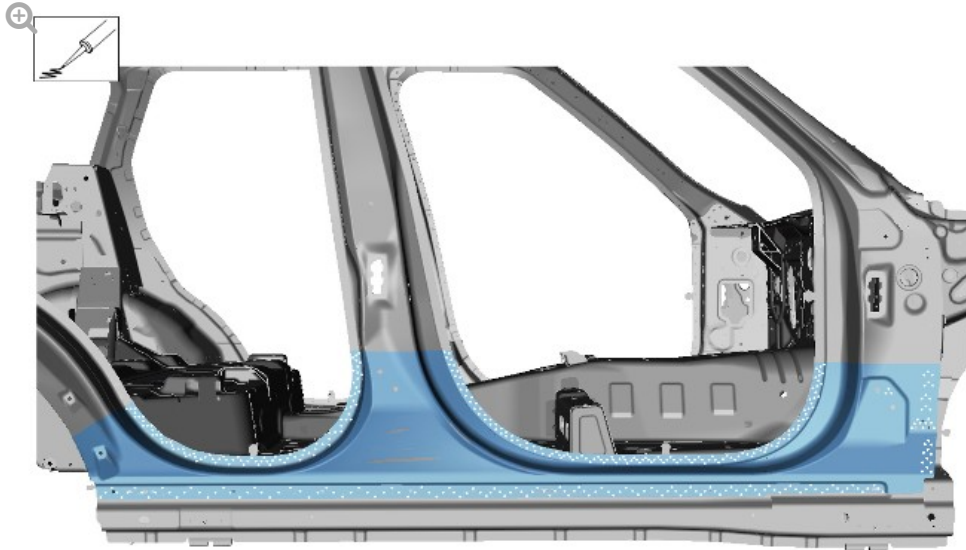
E143377

Install the NVH components and apply semi-rigid sealer to the NVH components.

21.

NOTE:

Make sure a continuous bead of adhesive surrounds fixing holes. Do not apply adhesive in the vicinity of the MIG welds.

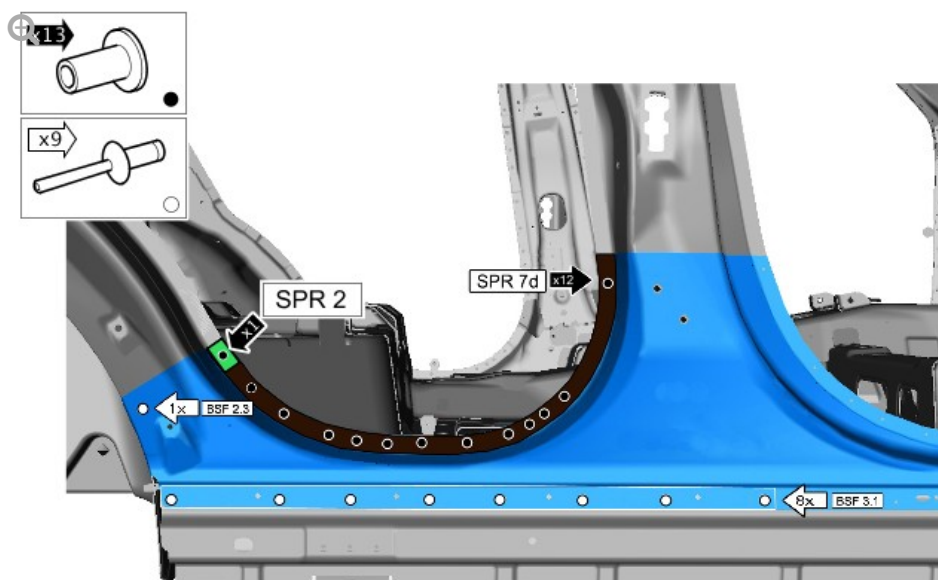


E144322

Apply a 5mm zig zag bead of the Land Rover recommended bonding material to the body joints.

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22. Offer up the new panel and clamp into position. Check alignment, if correct proceed to next step, if not, rectify and recheck before proceeding.
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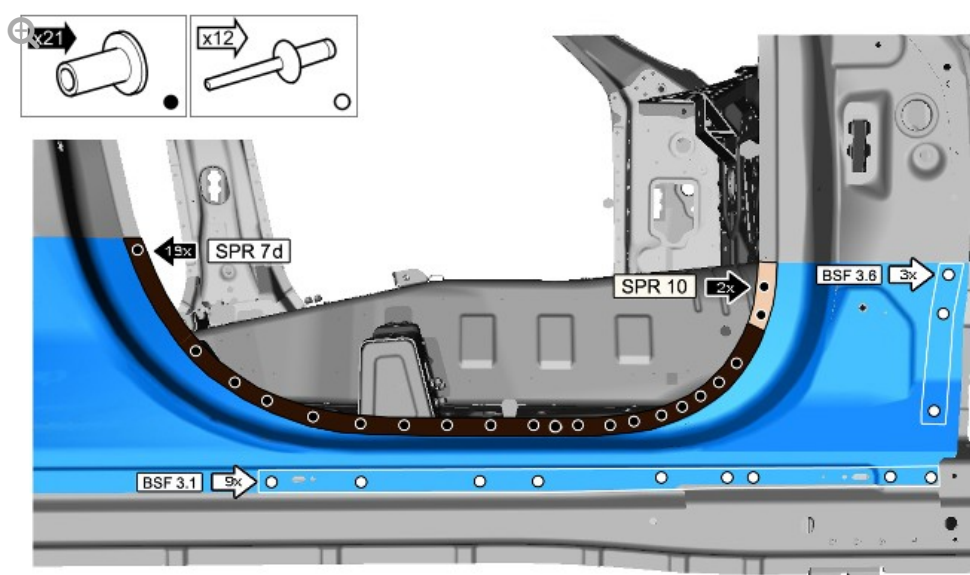
23.



E144323

Install the SPR's and BSF's at the points indicated.

24.



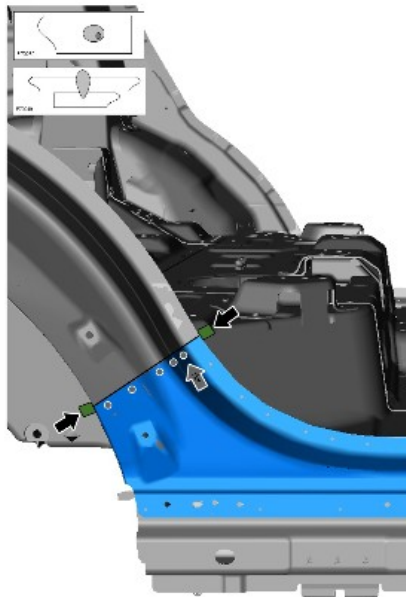
E144324

Install the SPR's and BSF's.

25. Remove any excess adhesive.

26. Tack weld the run-on/run-off tabs to the rocker panel.

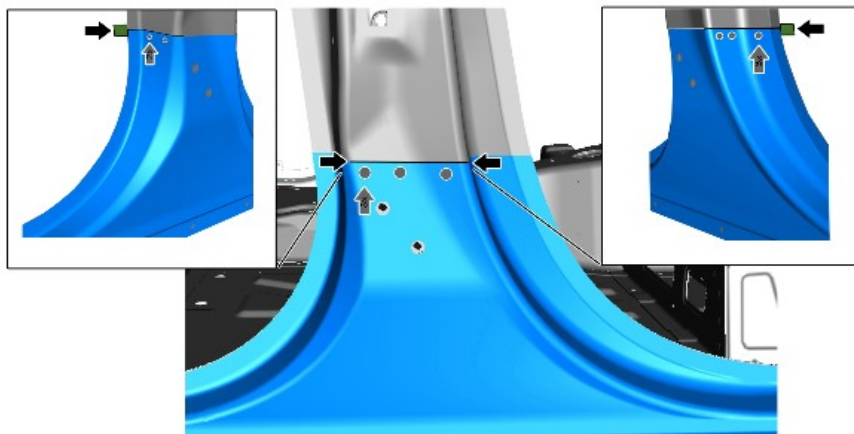
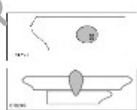
27.



E144325

MIG plug weld the backing plate as indicated and MIG weld the rocker panel butt joint.

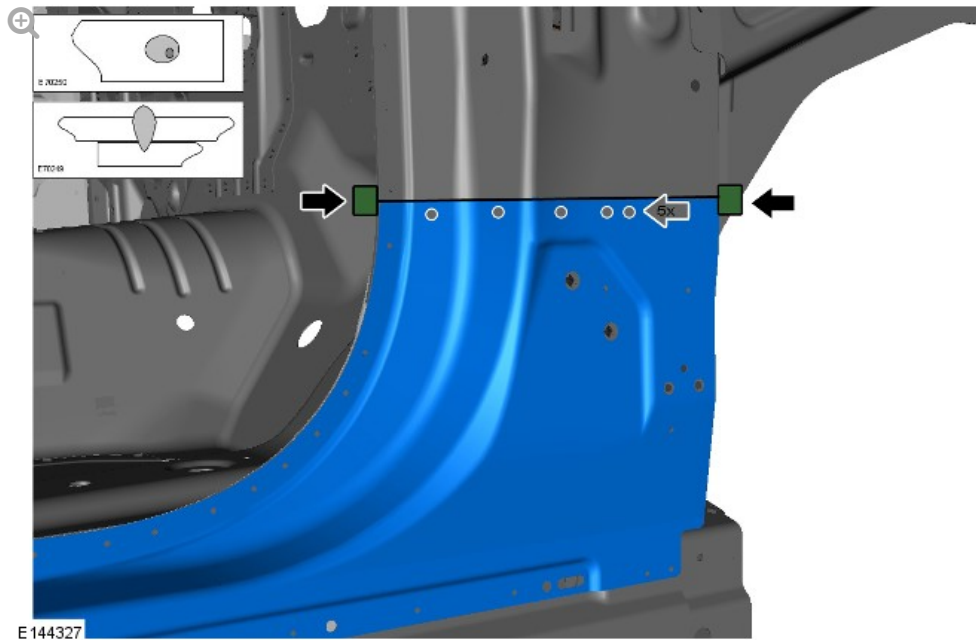
28.



E144326

MIG plug weld the backing plate as indicated and MIG weld the rocker panel butt joint.

29.



MIG plug weld the backing plate as indicated and MIG weld the rocker panel butt joint.

30. Cut off the run on/run off tabs.

31. Carry out a non destructive crack inspection on the MIG butt joints. If correct proceed to next step, if not, rectify and recheck before proceeding.

32. Dress the welded joints.

33. Make sure that any open or exposed panel joints are suitably sealed.

34. The installation of associated panels and components is the reversal of removal procedure.